Discrete Fourier Transform (DFT) — 01/09 | Monday

↳ numerically computable transform suitable for computer implementations

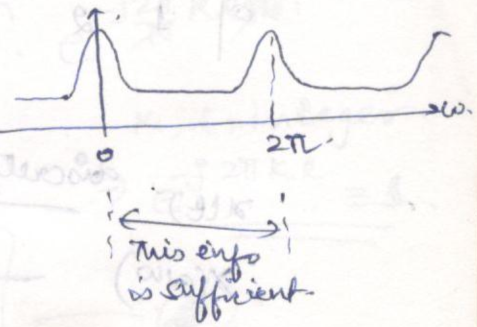
⇒ **Sampling of DTFT**

(sampling in frequency domain).

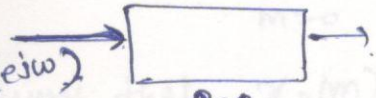
$x(n)$: Discrete time signal.

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n}$$

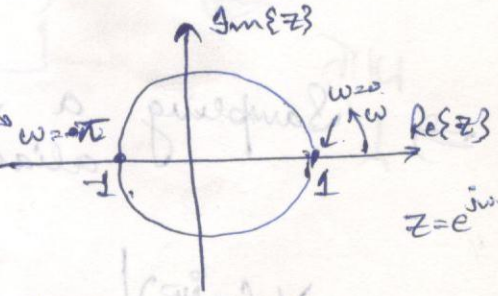
DTFT



Digital processor, accepts digital I/P only, so DTFT requires to be discretized.



$\omega = 2\pi$ & $\omega = 0$, values are same

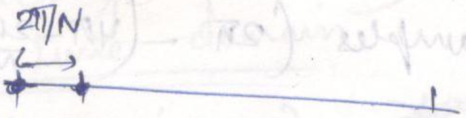


$z = e^{j\omega}$

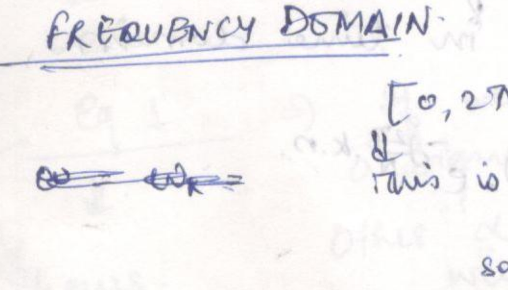
SAMPLING IN FREQUENCY DOMAIN

$[0, 2\pi)$

~~ω = ωk =~~ This is because, $\omega = 0$ & $\omega = 2\pi$, we have same values, so we don't count one of the sample.



If we want N samples ⇒ 1 sample width $= \left[\frac{2\pi}{N}\right]$.



$\therefore \omega = \omega_k = \frac{2\pi}{N} \cdot k, \quad 0 \leq k \leq N-1$

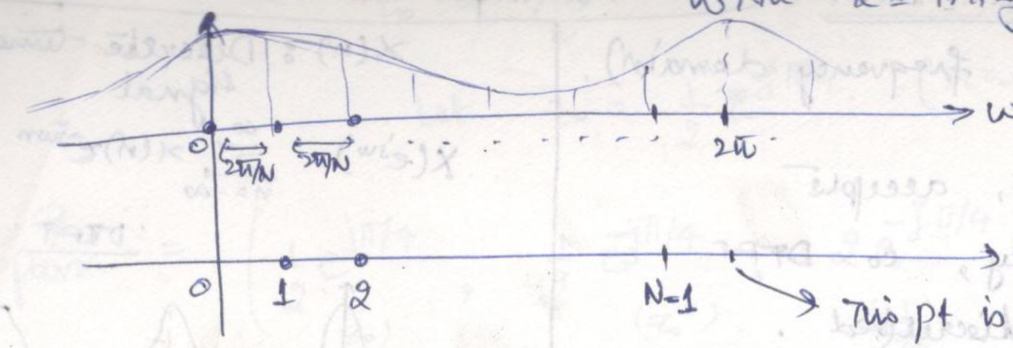
Suppose $N = 4$.

$\Rightarrow \omega = \omega_k = \frac{2\pi}{4} \cdot k, \quad 0 \leq k \leq 3$

$$\begin{bmatrix} k = 0 \Rightarrow \omega = 0 \\ k = 1 \Rightarrow \omega = \pi/2 \\ k = 2 \Rightarrow \omega = \pi \\ k = 3 \Rightarrow \omega = 3\pi/2 \end{bmatrix}$$

If we take 4 samples within a period of DTFT [0, 2π]

into k-axis
with k = integer



This pt is included in next period.

$x(t) \xrightarrow{\text{discretise}} x(n) = x(nT_s)$

$X(e^{j\omega}) \longrightarrow X(k) \quad .. \quad \omega = \frac{2\pi}{N} . k$

⇒ Sampling a signal in frequency domain ⇒ aliasing may occur in time domain.

$$X(e^{j\omega})\Big|_{\omega = \omega_k = \frac{2\pi}{N}.k} = \sum_{n=-\infty}^{\infty} x(n) . e^{-j\frac{2\pi}{N}.k.n}$$

$$\text{Samples}\left(0 - \left(2\pi - \frac{2\pi}{N}\right)\right) \equiv \text{samples}\left(2\pi - \left(4\pi - \frac{2\pi}{N}\right)\right)$$

$$\Rightarrow \quad X(e^{j\omega})\Big|_{\omega = \omega_k = \frac{2\pi}{N}.k} = \ldots + \sum_{n=-N}^{-1} x(n) e^{-j\frac{2\pi}{N}kn} + \sum_{n=0}^{N-1} x(n) * e^{-j\frac{2\pi}{N}kn} + \sum_{n=N}^{2N-1} x(n) e^{-j\frac{2\pi}{N}kn} + \ldots$$

$$\Rightarrow \quad = \sum_{l=-\infty}^{\infty} \sum_{n=lN}^{lN+N-1} x(n) e^{-j\frac{2\pi}{N}.kn}$$

for $l = 0$ — we get this summation

for $l = 1$ —

let $m = n - lN \Rightarrow n = m + lN$.

$\Rightarrow$ when $n = lN$ $\Rightarrow m = 0$. when $n = lN + N - 1$ $\Rightarrow m = N-1$.

$$\Rightarrow X\left(e^{-j\frac{2\pi}{N}k}\right) = \sum_{l=-\infty}^{\infty} \sum_{m=0}^{N-1} x(m+lN) \cdot e^{-j\frac{2\pi}{N} \cdot k \cdot (m+lN)}$$

$$\Rightarrow X(k) = \sum_{m=0}^{N-1} \sum_{l=-\infty}^{\infty} x(m+lN)\, e^{-j\frac{2\pi}{N}km} \cdot e^{-j\frac{2\pi}{\cancel{N}} kl\cancel{N}}$$

k, l = integer. $\Rightarrow e^{-j2\pi kl} = 1$.

Assume that $x_p(m) = \sum_{l=-\infty}^{\infty} x(m+lN)$. — (1)

$$\Rightarrow \boxed{X(k) = \sum_{m=0}^{N-1} x_p(m) \cdot e^{-j\frac{2\pi}{N}km}} \quad \text{— (2)}$$

eq 1 $\Rightarrow$ If we sample a signal in 1 domain, Other domain signal will be periodic.

⇓

Shows periodicity in time-domain

expanding eq (1) $\Rightarrow$ ~~$x_p(m)$~~ $x_p(n) = \sum_{l=-\infty}^{\infty} x(n+lN)$.

$x_p(n)$ $= \ldots + x(n-N) + x(n) + x(n+N) + \ldots$

$x(n-N)$: $(l=-1)$; $x(n)$: $(l=0)$; $x(n+N)$: $l=1$

Periodic extension of ~~$x(m)$~~ $x(n)$ [replacing m by n].

⇓

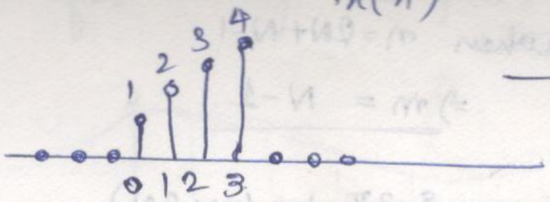
$x(n)$ is shifted right & left with period $\underline{N}$.

$x_p(n)$ is the periodic extention of $x(n)$, $\therefore$ $x(n)$ can be recovered form $x_p(n)$ if there is no aliasing in time domain.

Example

$x(n)$ length of signal $(x(n)) = L$

$\underline{L = 4}$ $\Longleftrightarrow$ no. of samples (L).



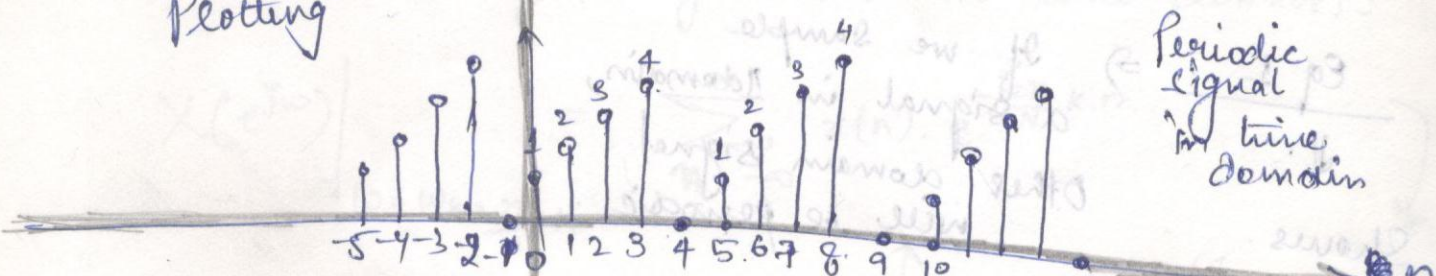
Case I. when $\underline{N > L}$.

N ⇓ no. of samples we are taking in 1 period of DTFT.

Suppose $N = 5$.

$\Rightarrow x_p(n) = \dots + x(n-5) + x(n) + x(n+5) + \dots$

⇓ Plotting



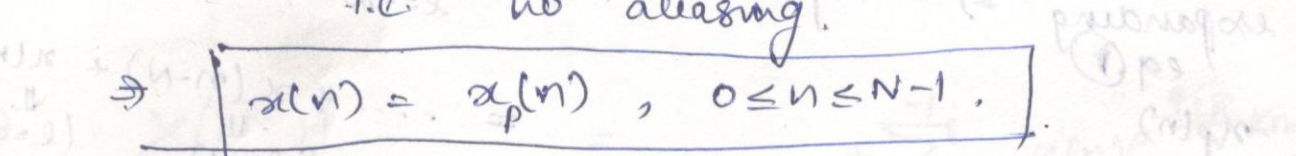
Periodic signal in time domain

we can ~~also~~ recover this signal.
i.e. no aliasing.

$$\Rightarrow \boxed{x(n) = x_p(n), \quad 0 \le n \le N-1.}$$

Case II. when $N = L$. $(\Rightarrow N = 4)$

$x_p(n) = \dots + x(n-4) + x(n) + x(n+4) + \dots$



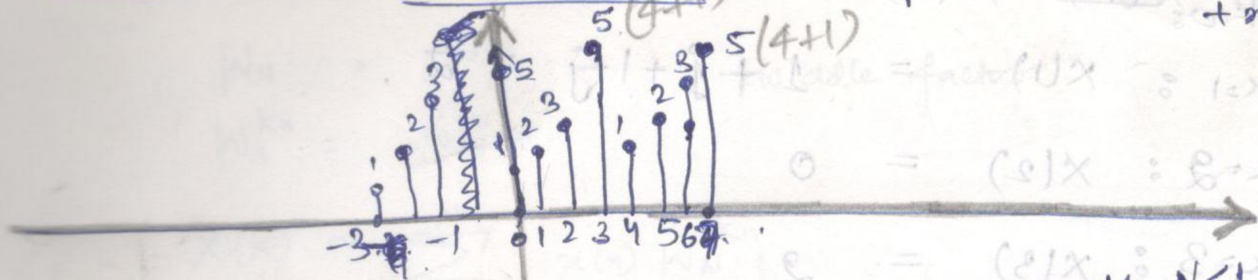
we can recover this signal

$$\boxed{x(n) = x_p(n), \quad 0 \le n \le N-1}$$

$\therefore$ for $N \ge L$, no aliasing occurs.

case III : $N < L$

$x_p(n) = \cdots + x(n-3) + x(n) + x(n+3) + \cdots$



Cant recover in case ~~[illegible]~~ $N < L$.

So

$\boxed{\text{Minimum point DFT must be greater or equal to no. of samples contained in } x(n).}$

From eq. ② ③.

$$X(K) = \sum_{n=0}^{N-1} x_p(n)\, e^{-j\frac{2\pi}{N}kn} \quad \{\text{replacing } m \text{ by } n\}.$$

$$= \sum_{n=0}^{N-1} x(n)\, e^{-j\frac{2\pi}{N}kn} \quad \{\text{considering anti-aliasing condition i.e. } N \geq L\}.$$

$0 \leq K \leq N-1$

$= N\text{-point DFT}$

N-point Inverse DFT

$$x(n) = \frac{1}{N} \sum_{k=0}^{N-1} X(k)\, e^{j\frac{2\pi}{N}kn}.$$

$0 \leq n \leq N-1.$

for example. $x(n) = \cos\left(n\frac{\pi}{2}\right)$, Compute 4-point DFT.

$\therefore$ max 4 samples in $x(n)$.

Sol.

$$x(n) = \{\underset{\uparrow}{1}, 0, -1, 0\}.$$

$0 \leq n \leq N-1 = 3.$

$$X(K) = \sum_{n=0}^{3} x(n)\, e^{-j\frac{2\pi}{4}\cdot k\cdot n}, \quad 0 \leq K \leq 3.$$

$\Rightarrow$ $K=0$: ~~x(k)~~ $X(0) = 1+0-1+0 = 0$.

$K=1$: $X(1) = 1+j+1-j = 2$.

$K=2$: $X(2) = 0$

$K=3$: $X(3) = 2$.

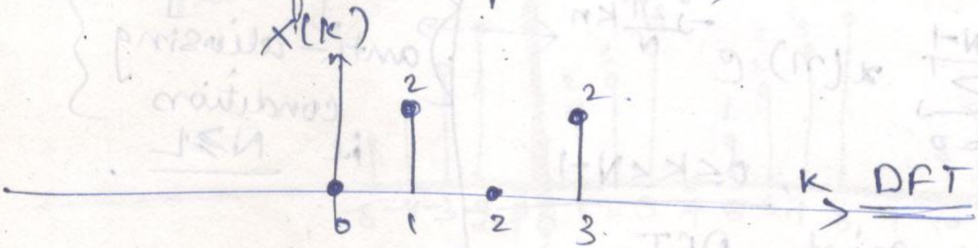
$$\Rightarrow X(K) = \{\underset{\uparrow}{0}, 2, 0, 2\}.$$

If we take inverse DFT, we get $x(n)$.

since $x(n) \longleftrightarrow X(k)$.

K is called frequency bin / index.

since if we plot it, we only show on integers, not on frequencies.



$W_N = e^{-j\frac{2\pi}{N}}$: twiddle factor

$W_N^{kn} = e^{-j\frac{2\pi}{N}\cdot kn}$.

$$X(K) = \sum_{n=0}^{N-1} x(n) \cdot W_N^{-kn}$$

Properties of Twiddle factor

① $W_N^N = 1$

② $W_N^{N/2} = -1$

③ $W_N^{N/4} = -j$

④ $W_N^{3N/4} = j$

⑤ $W_N^{K+N} = W_N^K$ (periodicity)

⑥ $W_N^{K+N/2} = e^{-j\frac{2\pi}{N}(K+\frac{N}{2})} = \underbrace{e^{-j\frac{2\pi}{N}K}}_{W_N^K} + \underbrace{e^{-j\frac{2\pi}{N}\cdot\frac{N}{2}}}_{=-1}$

$= -W_N^K$

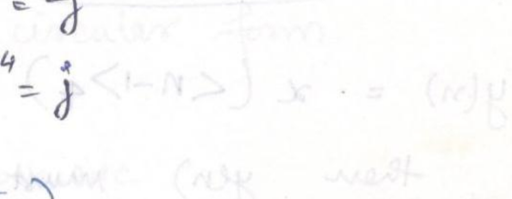
⑦ $W_N^{2K} = W_{N/2}^K$

⑧ $W_N^* = W_N^{-1}$

(skip here and explain with properties

~~Properties of DFT~~ Some Imp. Definitions.

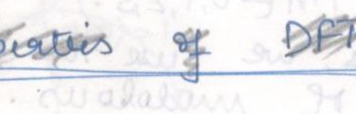
Circular shifting : $x(n)$:



or Periodic shifting

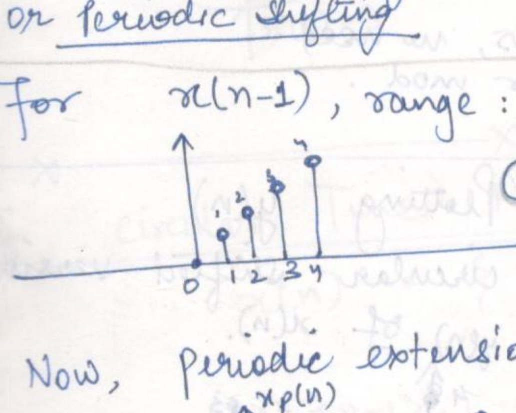
range $n : (0, 3)$

For $x(n-1)$, range : $(1, N)$

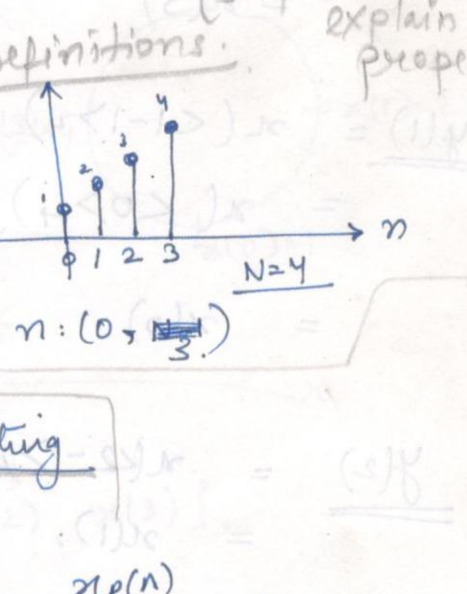


$\in$ Linear shifting

Now, periodic extension of $x(n)$ is $x_p(n)$



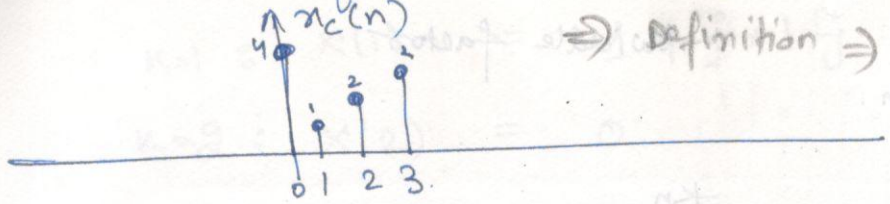
$x_p(n) = x(n)$



we want that after shifting $x(n)$ remains within the

$\langle\ \rangle$ modulo

circular shifted version of $x(n)$ is $x_c(n)$



$\Rightarrow$ Definition $\Rightarrow$ $x_c(n) = x(\langle n-1\rangle_N)$

$= \cancel{x((n-1)_N)}$

$= x(n-1, \bmod N)$.

1	2	3	4
4	1	2	3.

circular shifting

example.

$y(n) = x(\langle n-1\rangle_4)$, for $x(n)$, in $0 \le n \le 3$.

then $y(n)$ must also be in $0 \le n \le 3$.

$\underline{y(0)} = x(\langle 0-1\rangle_4) = x(\langle -1\rangle_4)$.

$\Downarrow$

But this index is outside our range

$(n \in 0,1,2,3.)$

Thus, we use the concept of modolous.

$\Rightarrow y(0) = x(-1+4)$

$= x(3)$

$\underline{y(1)} = x(\langle 1-1\rangle_4)$

$= x(\langle 0\rangle_4) \rightarrow$ Here it is in range, thus, no need for mod.

$= x(0)$

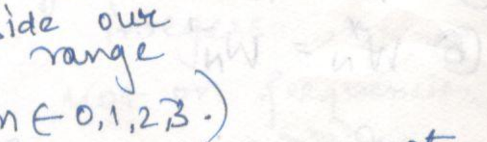
$\underline{y(2)} = x(\langle 2-1\rangle_4)$

$= x(1)$.

$y(3) = x(2)$.

Plotting $y(n)$

i.e. circular shifted version of $x(n)$.



✱ If for $x(\underbrace{\langle n-1\rangle}_{}{}_N)$

$\Downarrow$

is not within the range $n \in [0, N-1]$

then add/subtract N.

In general

for range of $x(n)$ is $[0, N-1]$.

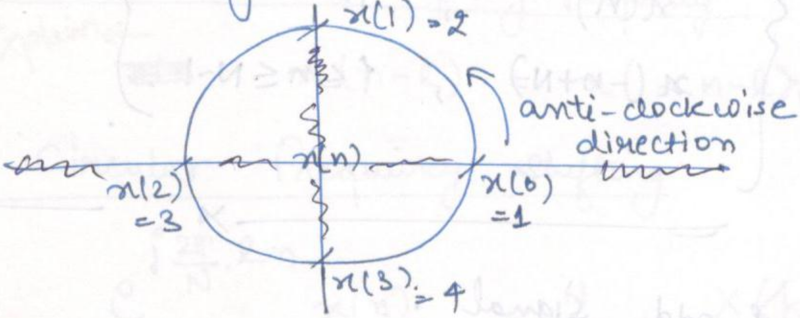
~~$x(n-l$~~

$$x(\langle n-l\rangle_N) = \begin{cases} x(n-l), & l \le n \le N-1 \\ x(n-l+N), & 0 \le n \le l-1 \end{cases}$$

$\cancel{n-l < N-1}$
$\cancel{n < N-1+l}$

x ———————— x ———————— x

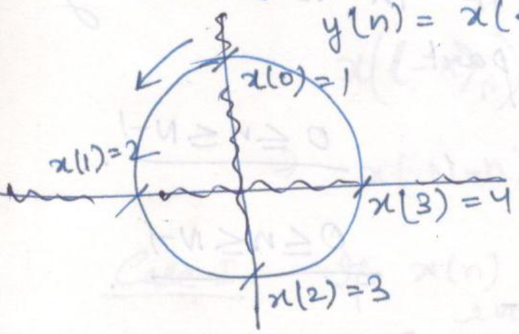
Plotting $x(n)$ in circular form.



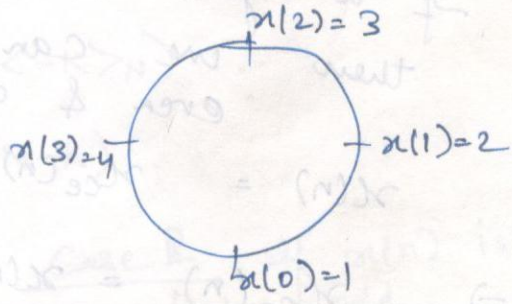
⇓

Shifted by (1)

$y(n) = x(\langle n-1\rangle_4)$



$y(n) = x(\langle n+1\rangle_4)$



x ———————— x ———————— x

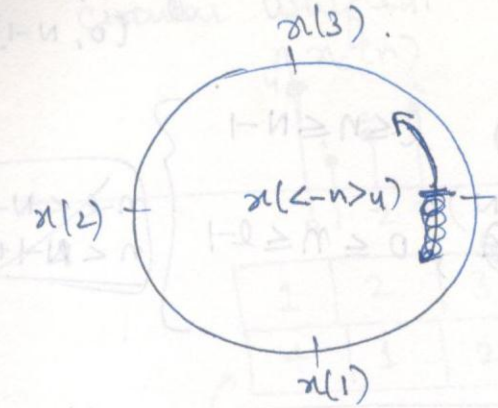
Circular Time Reversal.

$x(n) = \{ \underset{\uparrow}{x(0)}, x(1), x(2), x(3) \}$.

$x(\langle n-1\rangle_4) = \{ \underset{\uparrow}{x(3)}, x(0), x(1), x(2) \}$.

$x(\langle n+1\rangle_4) = \{ \underset{\uparrow}{x(1)}, x(2), x(3), x(0) \}$.

$x(\langle -n\rangle_4)$ ~~$x(-n)$~~ $= \{ \underset{\uparrow}{x(0)}, x(3), x(2), x(1) \}$.



$$x(\langle -n\rangle_N) = \begin{cases} x(n) & , n=0 \\ x(-n+N) & , 1 \le n \le N-1 \end{cases}$$

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circularly even & odd signal.

If a signal is neither ~~circularly~~ odd & even. then we can decompose it in its circularly even & circularly odd part.

$$x(n) = x_{ce}(n) + x_{co}(n) \quad , \quad 0 \le n \le N-1$$

$$\Rightarrow \quad x_{ce}(n) = \frac{x(n) + x(\langle -n\rangle_N)}{2} \quad , \quad 0 \le n \le N-1$$

and $$x_{co}(n) = \frac{x(n) - x(\langle -n\rangle_N)}{2} , \quad 0 \le n \le N-1$$

(If the signal is real, then only)

whenever we change the index which can go beyond the range $(0, N-1)$, it must be written in mod $\langle\ \rangle_N$ form.

If signal is not real.

then $x(n) \rightarrow$ circularly symmetric + circularly anti-symmetric.

DFT Properties Proof.

1. **Linearity** : $DFT[a x_1(n) + b x_2(n)]$

$$= \sum_{n=0}^{N-1} [a x_1(n) + b x_2(n)] W_N^{kn}$$

$$= a \sum_{n=0}^{N-1} x_1(n) W_N^{kn} + b \sum_{n=0}^{N-1} x_2(n) W_N^{kn}$$

$$= a X_1(k) + b X_2(k).$$

2. **Circular Time shifting** : $DFT[x(\langle n-l \rangle_N)]$

$$= \sum_{n=0}^{N-1} x(\langle n-l \rangle_N) W_N^{kn}$$

Breaking the limits according to time shifting of signal formula

DFT : $$= \sum_{n=0}^{l-1} x(n-l+N) W_N^{kn} + \sum_{n=l}^{N-1} x(n-l) W_N^{kn}$$

For 1st part : Let $(n-l+N) = r \Rightarrow n = r+l-N$

when $n=0 \Rightarrow r = N-l$

when $n = l-1 \Rightarrow r = N-1$

First part

$$\sum_{n=0}^{l-1} x(n-l+N) W_N^{kn} = \sum_{r=N-l}^{N-1} x(r) e^{-j\frac{2\pi}{N}k(r+l-N)}.$$

$$= \sum_{r=N-l}^{N-1} x(r) e^{-j\frac{2\pi}{N}k(r+l)} \cdot \underbrace{e^{j\frac{2\pi}{N}kN}}_{=1}$$

$$= \sum_{r=N-l}^{N-1} x(r) e^{-j\frac{2\pi}{N}k(r+l)}$$

Second Part : Let $n-l = r \Rightarrow n = r+l$

$\therefore$ when $n = l \Rightarrow r = 0$

when $n = N-1 \Rightarrow r = N-l-1$.

$\therefore$ ~~Second~~ $$\sum_{n=l}^{N-1} x(n-l) W_N^{kn} = \sum_{r=0}^{N-l-1} x(r) e^{-j\frac{2\pi}{N}k(r+l)}$$

Combining both ~~Terms~~ Parts :

$$DFT\left[x(\langle n-l\rangle_N)\right] = \sum_{r=N-l}^{N-1} x(r)\, e^{-j\frac{2\pi}{N}k(r+l)} + \sum_{r=0}^{N-l-1} x(r)\, e^{-j\frac{2\pi}{N}k(r+l)}$$

$$= \sum_{r=0}^{N-1} x(r)\, e^{-j\frac{2\pi}{N}k(r+l)}$$

$$= \sum_{r=0}^{N-1} x(r)\, e^{-j\frac{2\pi}{N}kr} \cdot e^{-j\frac{2\pi}{N}kl}$$

$$= X(k) \cdot e^{-j\frac{2\pi}{N}kl} = X(k)\, W_N^{kl}.$$

4. <u>Circular Time Reversal.</u>

$$DFT\left[x(\langle -n\rangle_N)\right] = \sum_{n=0}^{N-1} x(\langle -n\rangle_N)\, e^{-j\frac{2\pi}{N}kn}.$$

using Time reversal of signal formula.

$$= x(0) + \sum_{n=1}^{N-1} x(-n+N)\, e^{-j\frac{2\pi}{N}kn}$$

considering second part

Let $r = -n+N \Rightarrow n = N-r$

when $n=1$, $r = N-1$

$n = N-1$, $r = 1$.

$$\Rightarrow DFT\left[x(\langle -n\rangle_N)\right] = x(0) + \sum_{r=1}^{N-1} x(r)\, e^{-j\frac{2\pi}{N}k(-r+N)}$$

$$= x(0) + \sum_{r=1}^{N-1} x(r)\, e^{+j\frac{2\pi}{N}kr}$$

$$= \sum_{r=0}^{N-1} x(r)\, e^{j\frac{2\pi}{N}kr}$$

$$= \sum_{r=0}^{N-1} x(r)\, e^{-j\frac{2\pi}{N}(-k)r}$$

$$= X(\langle -k\rangle_N) = X(N-k).$$

Properties of DFT :

$x(n) \xleftrightarrow[DFT]{N} X(K)$: DFT pair.

I. Linearity $x_1(n) \xleftrightarrow{N} X_1(K)$

$x_2(n) \xleftrightarrow{N} X_2(K)$.

$\Rightarrow a x_1(n) + b x_2(n) \xleftrightarrow{N} a X_1(K) + b X_2(K)$.

2. Circular - Time shifting Property.

← ← Explain ←

$x(n-l) = x(\langle n-l \rangle_N) \xleftrightarrow{N} X(K).e^{-j\frac{2\pi}{N}.k.l}$

$= X(K) W_N^{Kl}$

3. Circular frequency shifting.

$e^{j\frac{2\pi}{N}.l.n} x(n) \xleftrightarrow{N} X(\langle K-l \rangle_N)$.

$= W_N^{-ln} x(n)$

4. Circular Time - Reversal.

$x(\langle -n \rangle_N) \xleftrightarrow{N} X(\langle -K \rangle_N)$.

$\Rightarrow x(N-n) \xleftrightarrow{N} X(N-K)$.

Case I. If $x(n)$ is circularly even.

$x(n) = x(\langle -n \rangle_N)$.

$= x(N-n)$.

then $X(K) = X(N-K)$.

ie. DFT : circularly even.

Case II If $x(n)$ is circularly odd.

$x(n) = -x(\langle -n \rangle_N)$.

$= -x(N-n)$

$\Rightarrow X(K) = -X(N-n)$.

DFT : circularly odd.

5. Conjugation - (or, conjugate Symmetry properties) $\boxed{Imp}$.

$x^*(n) \xleftrightarrow{N} X^*(\langle -K \rangle_N) = X^*(N-K)$ → → PTO

Case III $x(n)$ is real & circularly even.

$\Rightarrow x(n) = x^*(n) = x(\langle -n \rangle_N)$.

Therefore, $X(K) = X^*(\langle -K \rangle_N) = X(N-K) = X^*(N-K)$.

[illegible] DFT is also real & even.

case II. $x(n)$ is real & ~~circul~~

$$x(n) = x^*(n)$$

$$X(K) = X^*(\langle -K\rangle_N) = X^*(N-K).$$

DFT is conjugate symmetric.

Case III $x(n)$ is real & circularly odd.

$$x(n) = x^*(n) = -x(\langle -n\rangle_N).$$

$$\Rightarrow X(K) = X^*(\langle -K\rangle_N) = -X(\langle -K\rangle_N).$$

DFT is imaginary ~~real~~ & odd

Let

$x = jb$

$x^* = -jb$

$\Rightarrow x^* = -x.$

case IV. $x(n)$ is imaginary.

$$x(n) = -x^*(n).$$

$$\Rightarrow X(K) = -X^*(\langle -K\rangle_N)$$

DFT : conjugate anti-symmetric.

Case V. $x(n)$ is imaginary & even

$$x(n) = -x^*(n) = x(\langle -n\rangle_N).$$

$$X(K) = -X^*(\langle -K\rangle_N) = X(\langle -K\rangle_N)$$

DFT is imaginary & even

Case VI. $x(n)$ is imaginary & odd.

$$x(n) = -x^*(n) = -x(\langle -n\rangle_N)$$

$$X(K) = -X^*(\langle -K\rangle_N) = -X(\langle -K\rangle_N).$$

DFT is real & odd.

Summary.

$$x(n) = x_r^e(n) + x_r^o(n) + x_i^e(n) + x_i^o(n).$$

$$X(K) = X_r^e(K) + X_r^o(n) + jX_i^e(n) + jX_i^o(n).$$

in case of real, DFT properties are same.

but in case of imaginary, DFT ~~real~~ prop. interchange to even & odd.

04/09 | Thursday.

Case VII. $x(n)$ is neither even nor odd

i.e. it can be decomposed in its circulary even & odd parts.

Assuming $x(n) \Rightarrow$ real.

$$\Rightarrow x(n) = x_{ce}(n) + x_{co}(n).$$

We know, $x_{ce}(n) = \dfrac{x(n) + x(\langle -n\rangle_N)}{2}.$

$$\Rightarrow \text{DFT}\{x_{ce}(n)\} = \frac{1}{2}\left[X(K) + X(\langle -K\rangle_N)\right].$$

$x(n) \Rightarrow$ real, $= \dfrac{1}{2}\left[X(K) + X^*(K)\right].$ ∵ case I. (one) conjugate symmetric

$$= \frac{1}{2} \cdot 2 \cdot \text{Re}\{X(K)\} = X_R(K)$$

$Z = a + jb$

$Z^* = a - jb$

$Z + Z^* = 2a = 2\,\text{Re}\{Z\}.$

$Z - Z^* = 2jb = j2\,\text{Im}\{Z\}$

$$\Rightarrow \boxed{x_{ce}(n) \longleftrightarrow \text{Re}\{X(K)\} = X_R(k)}$$

circulary even part of $x(n)$ makes a pair with Real part of $X(k)$.

Now Since

$\Rightarrow x_{ce}(n) = x_{ce}(\langle -n\rangle)_N$

$$\Rightarrow \boxed{x_{ce}(\langle -n\rangle_N) \longleftrightarrow X_R(k)}$$

Case II. (real & even)

Also

$$\boxed{X_R(k) = X_R(\langle -k\rangle_N)}$$

i.e. Real part of DFT is always even

$\Rightarrow$

Now

$$x_{co}(n) = \frac{x(n) - x(\langle -n \rangle_N)}{2}$$

$$DFT\{x_{co}(n)\} = \frac{1}{2}\left[X(k) - X(\langle -k \rangle_N)\right].$$

from case I :

$$= \frac{1}{2}\left[X(k) - X^*(k)\right]$$

$$= j \cdot \frac{1}{2} \cdot 2\, Im\{X(k)\}.$$

$$= j\, Im\{X(k)\}. = j \cdot X_I(k).$$

$$\Rightarrow \boxed{x_{co}(n) \longleftrightarrow j \cdot X_I(k)}$$

Since its odd. $\Rightarrow x_{co}(n) = -x_{co}(\langle -n \rangle_N).$

$$\Rightarrow j X_I(k) = -j X_I(\langle -n \rangle_N)$$

$$\Rightarrow \boxed{X_I(k) = -X_I(\langle -n \rangle_N)}$$

Imaginary part of DFT is always odd.

Summary: $\Rightarrow$ $x(n)$ ... real.

$$x(n) \xleftrightarrow{N} X(k).$$

$X(k) = X_R(k) + X_I(k)$

complex

$x(n) = x_{ce}(n) + x_{co}(n).$

and $x(n)$ real $\Rightarrow$

$$x_{ce}(n) \xleftrightarrow{N} X_R(k) \rightarrow \text{real part of DFT is even}$$

$$x_{co}(n) \xleftrightarrow{N} j \cdot X_I(k) \rightarrow \text{Imaginary part of DFT is odd.}$$

for a real signal

$\Rightarrow$ Magnitude Spectrum is always even. $\rightarrow$ ~~proof~~ is above

$\Rightarrow$ phase " " " odd.

6. Circular Convolution / Periodic convolution.

$$x_1(n) \xleftrightarrow{N} X_1(k).$$

$$x_2(n) \xleftrightarrow{N} X_2(k).$$

Linear convolut^n

$x_1(n) * x_2(n)$

$= x_3(n)$

$= \sum_{m=0}^{L-1} x_1(m) x_2(n-m)$

$L = N_{x_1} + N_{x_2} - 1$

$$\underbrace{x_1(n) \circledast x_2(n)}_{= x_3(n)} \xleftrightarrow{N} \underbrace{X_1(k) \cdot X_2(k)}$$

$$\xleftrightarrow{N.} X_3(k).$$

MUST

$$\boxed{, 0 \le n \le N-1.}$$

$$\Rightarrow x_3(n) = x_1(n) \circledast x_2(n)$$

$$= x_1(n) \; N \; x_2(N.$$

→ N-point circular convolution.

i.e. if $x_1(n)$ contains 4 samples

& $x_2(n)$ " " "

then $x_3(n)$ must contain 4 samples. (which is not ~~the~~ same as in case of Linear convolution)

Formula for convolution

$$\boxed{x_3(n) = \sum_{m=0}^{N-1} x_1(m) \, x_2(\langle n-m \rangle_N).}$$

↑ circular convolution ↑.

→ mod N done due to change in index.

Proof.

$$X_3(k) = \sum_{n=0}^{N-1} x_3(n) \cdot e^{-j\frac{2\pi}{N}kn}$$

$$= \sum_{n=0}^{N-1} \left[x_1(n) \circledast x_2(n) \right] e^{-j\frac{2\pi}{N}kn}$$

$$= \sum_{n=0}^{N-1} \sum_{m=0}^{N-1} x_1(m) \cdot x_2(\langle n-m \rangle_N) \, e^{-j\frac{2\pi}{N}kn}$$

$$= \sum_{m=0}^{N-1} x_1(m) \underbrace{\sum_{n=0}^{N-1} x_2(\langle n-m \rangle_N) \, e^{-j\frac{2\pi}{N}kn}}$$

→ Time Shifting DFT

$$= \sum_{m} x_1(m) \; X_2(k) \cdot e^{-j\frac{2\pi}{N}km}$$

$$\boxed{X_3(k) = X_1(k) \cdot X_2(k).}$$

3 Methods

Analytical, ~~Mathematical~~ Matrix, graphical Method for solving circular convolution.

$\boxed{\text{ANALYTICAL}}$ (using formula)

example.

$$x_1(n) = \{ \underset{\uparrow}{1}, 2, 3, 4 \}.$$

$$x_2(n) = \{ \underset{\uparrow}{1}, 0, 0, 1 \}.$$

Perform circular convolution

$$x_3(n) = x_1(n) \; ④ \; x_2(n), \quad 0 \leq n \leq N-1.$$

$$\Rightarrow 0 \leq n \leq 3.$$

$$x_3(n) = \sum_{m=0}^{3} x_1(m) \cdot x_2(\langle n-m \rangle_4)$$

$$x_3(0) = x_1(0) \cdot x_2(0) + x_1(1) \cdot x_2(3) + x_1(2) \cdot x_2(2) + x_1(3) \cdot x_2(1) \quad \ldots (1)$$

$$= 1 + 2 + 0 + 0 = 3.$$

$$x_3(1) = x_1(0)\, x_2(1) + x_1(1)\, x_2(0) + x_1(2) \cdot x_2(3) + x_1(3) \cdot x_2(2) \quad \ldots (2)$$

$$= 0 + 2 + 3 = 5.$$

$$\boxed{\Rightarrow \quad x_3(n) = \{ \underset{\uparrow}{3}, 5, \cdot, \cdot \}}$$

$$x_3(2) = x_1(0)\, x_2(2) + x_1(1)\, x_2(1) + x_1(2) \cdot x_2(0) + x_1(3) \cdot x_2(3) \quad \ldots (3)$$

$$= 0 + 0 + 3 + 4 = 7$$

$$x_3(3) = x_1(0)\, x_2(3) + x_1(1)\, x_2(2) + x_1(2)\, x_2(1) + x_1(3) \cdot x_2(0)$$

$$= 1 + 0 + 0 + 4 = 5$$

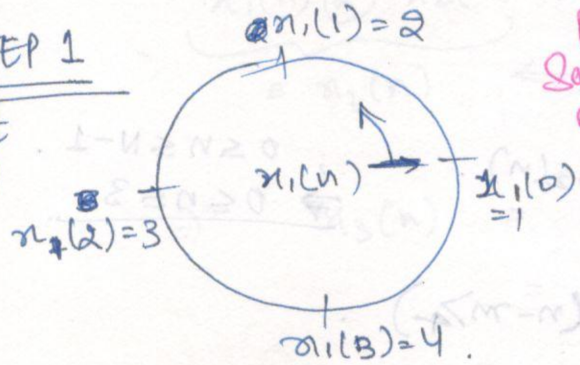
GRAPHICAL

$$x_3(n) = \sum_{m=0}^{3} x_1(m) \cdot x_2(\langle n-m \rangle_4) \quad , \; 0 \le n \le 3.$$

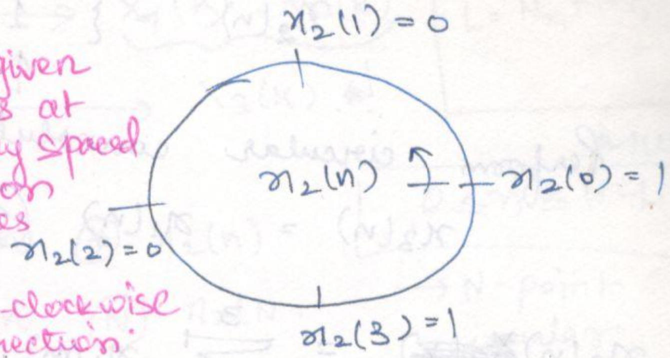
$$\boxed{n=0 \; ; \; x_3(0) = \sum_{m=0}^{3} x_1(m) \cdot x_2(\langle -m \rangle_4).}$$

STEP 1

Plot $x(n)$

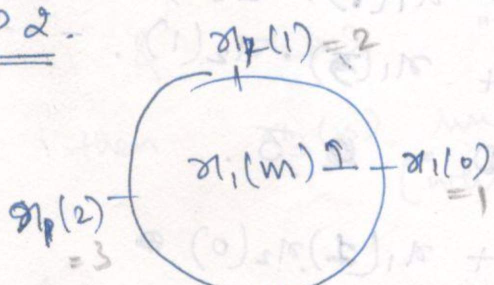


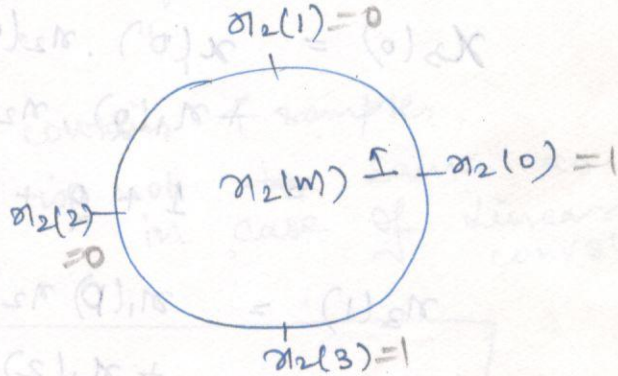
Plot given sequences at equally spaced pts on circles in anti-clockwise direction.



STEP 2.

change index



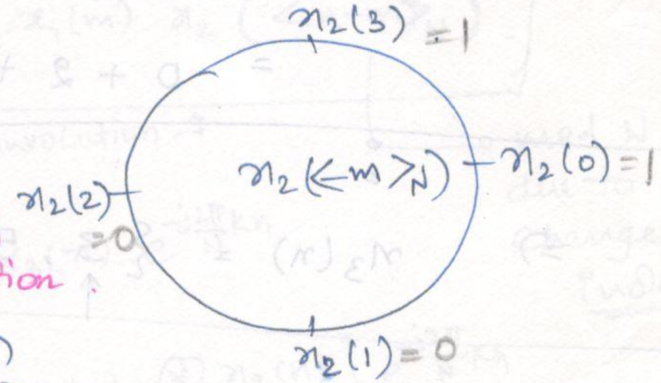


STEP 3.

we need $x_2(\langle -m \rangle_N)$.

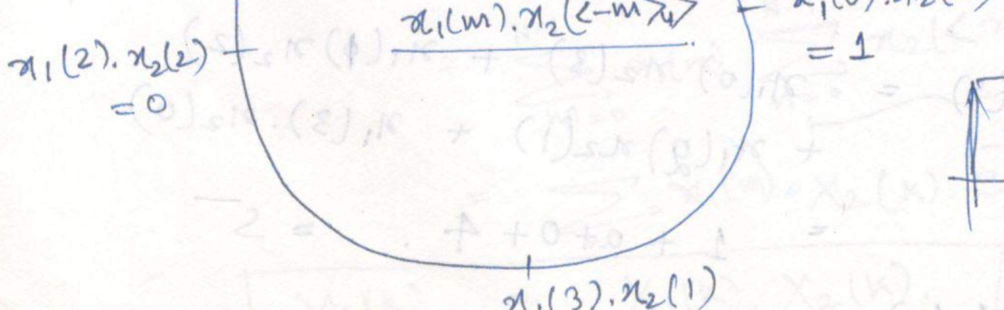
Fold one of the sequence to get $x(\langle -m \rangle_N)$

i.e. represent $x(m)$ in clockwise direction.



Step 4.

Multiply & Add



Add.

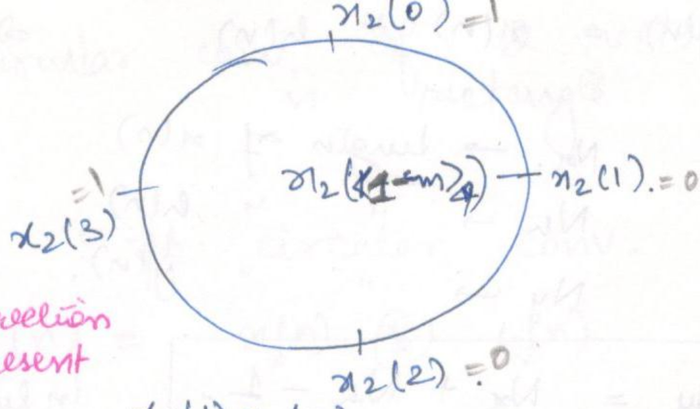
$$\boxed{x_3(0) = 3}$$

$$\boxed{\text{for } n = 1; \quad x_3(1) = \sum_{m=0}^{3} x_1(m)\, x_2(\langle 1-m \rangle_4).}$$

Step 1. Thus, only time shift $x_2(\langle -m \rangle_4)$ by 1.

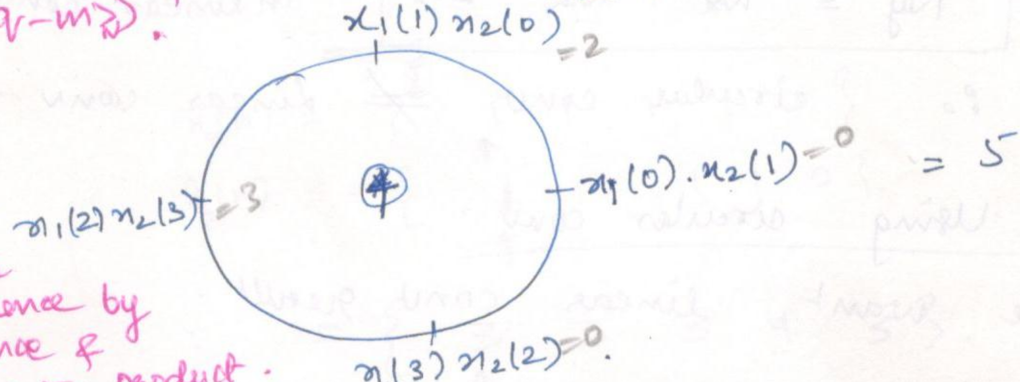
we already have $x_1(m)$. & $x_2(\langle -m \rangle_4)$.

rotating anti-clockwise $x(\langle -m \rangle_4)$.



- Rotate the sequence $x(\langle -m \rangle_N)$ in anticlockwise direction "q" times to represent $x(\langle q-m \rangle_N)$.

Step 2. multiply & sum.



$= 5$.

Multiply the rotated sequence by other sequence & finally add the product.

$\boxed{\text{for } n = 2; \quad \text{rotate } x(\langle 1-m \rangle_4) \text{ by } 1.}$ anti-clockwise method

Repeat this procedure for all possible values of n, $0 \le n \le N-1$ to get final sequence $x_3(n)$.

Matrix method.

$x_1(n) = \{1, 2, 3, 4\}$

$x_2(n) = \{1, 0, 0, 1\}$.

4-point circular convolution is:

$$\begin{bmatrix} x_3(0) \\ x_3(1) \\ x_3(2) \\ x_3(3) \end{bmatrix} = \begin{bmatrix} x_2(0) & x_2(3) & x_2(2) & x_2(1) \\ x_2(1) & x_2(0) & x_2(3) & x_2(2) \\ x_2(2) & x_2(1) & x_2(0) & x_2(3) \\ x_2(3) & x_2(2) & x_2(1) & x_2(0) \end{bmatrix} \begin{bmatrix} x_1(0) \\ x_1(1) \\ x_1(2) \\ x_1(3) \end{bmatrix}.$$

$$= \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \\ 7 \\ 5 \end{bmatrix}.$$

* Performing Linear Convolution using Circular convolution (or using DFT).

$$x(n) \longrightarrow \boxed{h(n)} \longrightarrow y(n).$$

$y(n) = x(n) * h(n)$ ⇒ response of LTI system is obtained using linear conv. only.

Let $N_x \rightarrow$ length of $x(n)$

$N_h \rightarrow$ " " $h(n)$

$N_y \rightarrow$ " " $y(n)$.

$$\boxed{N_y = N_x + N_h - 1.}$$ in linear conv.

∴ circular conv $\neq$ linear conv.

Using circular conv.

if we want linear conv result.

⇒ $x_z(n) = \{ x(n) \underbrace{0 . 0 . 0 . . 0}_{(N_y - N_x)\text{ zeros}} \}$

⇓

Zero padded version of $x(n)$ i.e. Making total length N_y

$h_z(n) = \{ h(n) \underbrace{0. 0. 0 . . . 0}_{(N_y - N_h)\text{ zeros}} \}$

Now $\boxed{y(n) = x_z(n) \circledast h_z(n).}$ Now result will be same as linear one

example.

$x(n) = \{ \underset{\uparrow}{1} \; 1 \; 1 \; 1 \}$ ... $N_x = 4$

$h(n) = \{ \underset{\uparrow}{1} \; 1 \; 1 \; 1 \}$ ... $N_h = 4$

⇒ So with linear conv

$N_y = 4 + 4 - 1 = 7$

$y(n) = \{ 1 \; 2 \; 3 \; 4 \; 3 \; 2 \; 1 \}$

$\Rightarrow$ 4-point circular conv. $(N=4)$

$y_1(n) = x(n) \; ④ \; h(n)$

$= \{ \underset{\uparrow}{4} \;\; 4 \;\; 4 \;\; 4 \}$.

i.e. circular conv. of 2 rectangles is rectangle.

$\Rightarrow$ Compute 5-pt. circular conv. $(N=5)$.

$y_2(n) = x_z(n) \; ⑤ \; h_z(n)$.

$x_z(n) = \{ \underset{\uparrow}{1} \;\; 1 \;\; 1 \;\; 1 \;\; 0 \}$

$h_z(n) = \{ \underset{\uparrow}{1} \;\; 1 \;\; 1 \;\; 1 \;\; 0 \}$

$= \{ \underset{\uparrow}{3} \;\; 3 \;\; 3 \;\; 4 \;\; 3 \}$.

$\Rightarrow$ Compute 6-pt C.C. $(N=6)$

$y_3(n) = x_z(n) \; ⑥ \; h_z(n)$.

$x_z(n) = \{ 1 \;\; 1 \;\; 1 \;\; 1 \;\; 0 \;\; 0 \}$.

$h_z(n) = \{ 1 \;\; 1 \;\; 1 \;\; 1 \;\; 0 \;\; 0 \}$.

$y_3(n) = \{ \underset{\uparrow}{2} \;\; 2 \;\; 3 \;\; 4 \;\; 3 \;\; 2 \}$.

$\Rightarrow$ Compute 7-pt. CC $(N=7)$

$y_4(n) = x_z(n) \; ⑦ \; h_z(n)$.

$= \{ \underset{\uparrow}{1} . \;\; 2 \;\; 3 \;\; 4 \;\; 3 \;\; 2 \;\; 1 \}$

* Circular convolution is linear conv. with aliasing

7-pt se 6-pt kaise calculate karien.

{ 1, 2, 3, 4, 3, 2, 1 }

add

6-pt. { 2, 2, 3, 4, 3, 2 }.

for 5-pt: { 1, 2, 3, 4, 3, 2, 1 }

add 2 add 1 1

{ (2+1) (2+1), 3, 4, 3 }

for 4pt { 1 2 3 4 3 2 1 }

add add add

II Duality Property

$$x(n) \xleftrightarrow[DFT]{N} X(k)$$

$$X(n) \xleftrightarrow[DFT]{N} N \cdot x(\langle -k \rangle_N)$$

further

$$DFT[x(n)] = X(k)$$ ⇒ change of variable

$$DFT\{DFT[x(n)]\} = DFT[X(k)]$$ taking DFT again

$$= N\, x(\langle -n \rangle_N)$$ ∴ ⇒ again change of variable

example: $x(n) = \{\underset{\uparrow}{1}, 1, 1, 1\}$

Take 4-pt DFT

DFT

$$X(k) = \{\underset{\uparrow}{4}, 0, 0, 0\}$$

Taking this as sequence

~~x(n)~~ Let $g(n) = \{\underset{\uparrow}{4}, 0, 0, 0\}$

Then Taking its DFT.

$$G(k) = \{\underset{\uparrow}{4}, 4, 4, 4\}$$

$$= N\, x(\langle -n \rangle_N)$$

Duality

$$= 4\, x(\langle -n \rangle_4)$$

$$DFT[\cdot] = \mathcal{F}[\cdot]$$

ex.

$$\mathcal{F}\left[\mathcal{F}\left[\underbrace{\mathcal{F}\left[\mathcal{F}[x(n)]\right]}_{\equiv N x(\langle -n \rangle_N)}\right]\right] = ?$$

use duality property

$N x(\langle -n \rangle_N)$ → let $g(n)$

Now compute ⇒ $\mathcal{F}[\mathcal{F}[g(n)]] = N \cdot g(\langle -n \rangle_N)$

⇒ final ans $= \boxed{N^2 x(n)}$ since twice time reversal in signal

8. Parseval's Theorem.

If $x(n) \xleftrightarrow{N} X(k)$

$$\boxed{\sum_{n=0}^{N-1} |x(n)|^2 = \frac{1}{N} \sum_{k=0}^{N-1} |X(k)|^2} \quad \text{Prove}$$

Proof   L.h.s.

$$\sum_{n=0}^{N-1} |x(n)|^2 = \sum_{n=0}^{N-1} x(n) \cdot x^*(n)$$

$\Downarrow$ Inverse DFT

$$= \sum_{n=0}^{N-1} x(n) \left[\frac{1}{N} \sum_{k=0}^{N-1} X(k) \cdot e^{j\frac{2\pi}{N}kn} \right]^*$$

$$= \sum_{n=0}^{N-1} x(n) \cdot \frac{1}{N} \sum_{k=0}^{N-1} X^*(k) \cdot e^{-j\frac{2\pi}{N}kn}$$

$\Downarrow$

$$= \frac{1}{N} \sum_{k=0}^{N-1} X(k) \cdot X^*(k)$$

$$= \frac{1}{N} \sum_{k=0}^{N-1} |X(k)|^2$$

ex. $x(n) = \delta(n)$

$$X(k) = \sum_{n=0}^{N-1} \delta(n) \cdot e^{-j\frac{2\pi}{N}kn}$$

$$= e^{-j\frac{2\pi}{N}kn} \Big|_{n=0} = 1$$

ex.
$$x(n) = \begin{cases} 1, & n = \text{even} \longrightarrow 0 \le n \le N-1 \\ 0, & n = \text{odd} \longrightarrow 0 \le n \le N-1 \end{cases}$$

Compute N-pt DFT

$$X_e(k) = \sum_{n=0}^{N-1} x_e(n) \cdot e^{-j\frac{2\pi}{N}kn}$$

$$= \sum_{r=0}^{\frac{N}{2}-1} x(2r) \cdot e^{-j\frac{2\pi}{N}k \cdot 2r}$$

$$+ \sum_{r=0}^{\frac{N}{2}-1} x(2r+1)\, e^{-j\frac{2\pi}{N}k(2r+1)}$$

even $n \rightarrow (2r) \Rightarrow n = 0$ to $\frac{N}{2}-1$ $\boxed{0-3}$ ∴ 4 values of r.

odd $n \rightarrow (2r+1)$ ~~⇒ ...~~

$$= \sum_{r=0}^{\frac{N}{2}-1} e^{-j\frac{2\pi}{N}k(2r)}$$

$$= \sum_{r=0}^{N/2-1} \left(e^{-j\frac{4\pi}{N}k}\right)^{r}$$

understanding ⇓

~~condition~~ considering $\displaystyle\sum_{r=0}^{N-1} \alpha^r = \begin{cases} N, & \alpha = 1 \\ \dfrac{1-\alpha^N}{1-\alpha}, & \alpha \neq 1 \end{cases}$

Now if $k = 0$, $\alpha = 1$

$k = \frac{N}{2}$, $\alpha = 1$ $\left(e^{-j2\pi} = 1\right)$

$$\Rightarrow \quad = \begin{cases} \dfrac{N}{2}, & k = 0, \frac{N}{2} \\ \dfrac{1 - e^{-j\frac{4\pi}{N}k\frac{N}{2}}}{1 - e^{-j\frac{4\pi}{N}k}} = 0 & \text{i.e. } 0 \text{ otherwise} \end{cases}$$